



- (d) Use the Maclaurin's series for e^x given in the list of formulae (MF19) to find the first three terms of the series expansion of $z(1 - e^{-\frac{1}{z}})$, in ascending powers of $\frac{1}{z}$, and deduce the value of $\lim_{n \rightarrow \infty} (U_n)$. [3]

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- 7 (a) Show that $\frac{d}{dx}(\ln(\tanh x)) = 2 \operatorname{cosech} 2x$. [3]

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